Genetic evidence supporting the fecal-perineal-urethral hypothesis in cystitis caused by Escherichia coli.

PURPOSE: The fecal-perineal-urethral hypothesis to explain the cause of urinary tract infections (UTI) by enteric bacteria has been supported by longitudinal studies using methods of serotyping and detecting urovirulence factors such as P fimbriae. However, genetic techniques to more accurately characterize Escherichia coli strains have not been exploited. MATERIALS AND METHODS: A total of 2,700 E. coli colonies isolated from the urine and rectal swabs of 9 female subjects with acute uncomplicated cystitis and from the rectal swabs of 30 healthy women were serotyped and examined for genes encoding various urovirulence factors by colony hybridization test. The clonality of the urine and fecal isolates of E. coli from the cystitis subjects was further evaluated by pulsed-field gel electrophoresis (PFGE). RESULTS: E. coli strains causing cystitis dominated the rectal flora of 7 of 9 patients. In the remaining 2 patients, similar clones comprised at least 20% of the fecal flora. Carriage of E. coli strains with a variety of urovirulence factors was quite common among healthy women. PFGE demonstrated that most of the isolates sharing the same serotypic characteristics and virulence factors in the urine and rectal swab samples from each subject were identical. CONCLUSIONS: Based upon precise genetic techniques, our results clearly support the fecal-perineal-urethral hypothesis, indicating that E. coli strains residing in the rectal flora serve as a reservoir for urinary tract infections, e.g., cystitis.